Practical 4 report
RNA-Seq: Differential gene expression analysis

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1 Abstract

RNA Sequencing (RNA Seq) is a next generation sequencing method used to identify and quantify all of the RNA molecules given in a biological sample. This method is a highly sensitive approach to measuring gene expression levels, enabling detection of both highly and lowly expressed transcripts. One of the most significant applications of RNA Seq is observing differential gene expression between two conditions, most commonly between healthy and diseased tissues. In this case, RNA-Seq analysis was performed using data obtained from patients in different stages of bile duct cancer. By comparing gene expression among patients in different stages, we can get a clearer view of which genes are included in disease progression and could potentially be therapeutic targets.

2 Methods

2.1 Data characteristics

Data were obtained from the Cancer Genome Atlas (TCGA), from Recount resource. These online sources provide publicly available genomic data that help cancer researchers develop new potential treatments. The data were given in a Range Summarized Experiment (RSE) format. It contained information about 45 cancer patients and their genes. None of the patients were excluded from the analysis because they all met the required conditions and their cancer stage was recorded.

2.2 Tools and Packages used

RStudio was used to perform the data analysis including some of its packages:

- DESeq2 - differential gene expression analysis
- biomaRt - gene annotation
- ClusterProfiler - post RNASeq enrichment analysis
- ggplot2, grepel - data visualization

Some additional packages used are dplyr for data manipulation, Summarized-Experiment that gives us the structure of genomic experiments and pathview, for KEGG pathway visualization with gene expression overlay.

2.3 Statistical Analysis

A log2 fold change was calculated for each gene to determine the intensity and the direction of expression. Adjusted p-values were obtained using false discovery rate (FDR) control and genes with value less than 0.001 were considered statistically significant. This low threshold was chosen to avoid false positive results because a large number of genes was tested simultaneously.

3 Results

3.1 Distribution of patients across cancer stages

Firstly, the patients were divided into categories based on their stage of cancer. There were 36 patients in the early stages and 9 in the late stages. There is a noticable difference between the number of patients in both groups considering that there are more late stages than early ones in the selected sample.

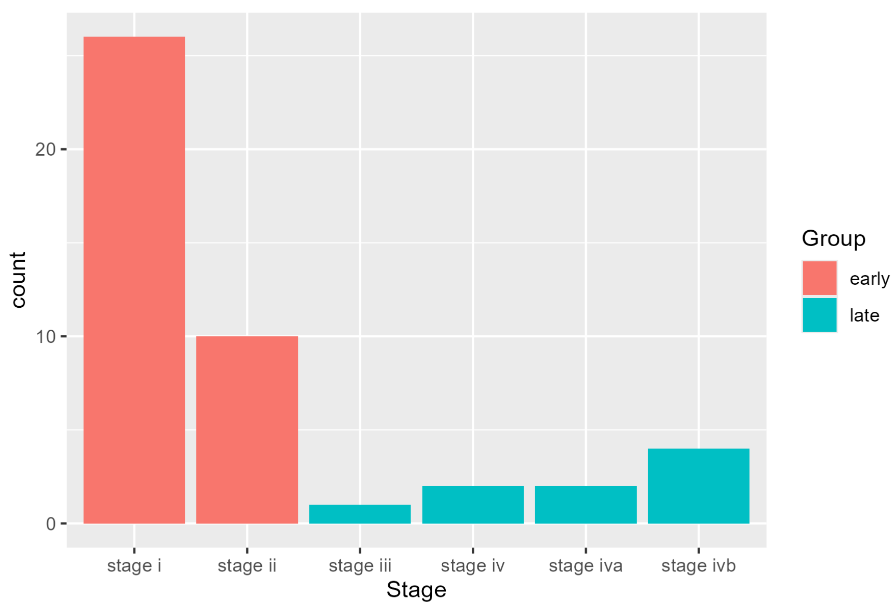

Figure 1: Bile duct cancer patients across stages

3.2 Differential gene expression

58,037 genes were analysed in total, however after excluding rows with NA values,49392 genes were left. Out of the remaining genes 224 were underexpressed,26 overexpressed and 49142 were not differentially expressed.

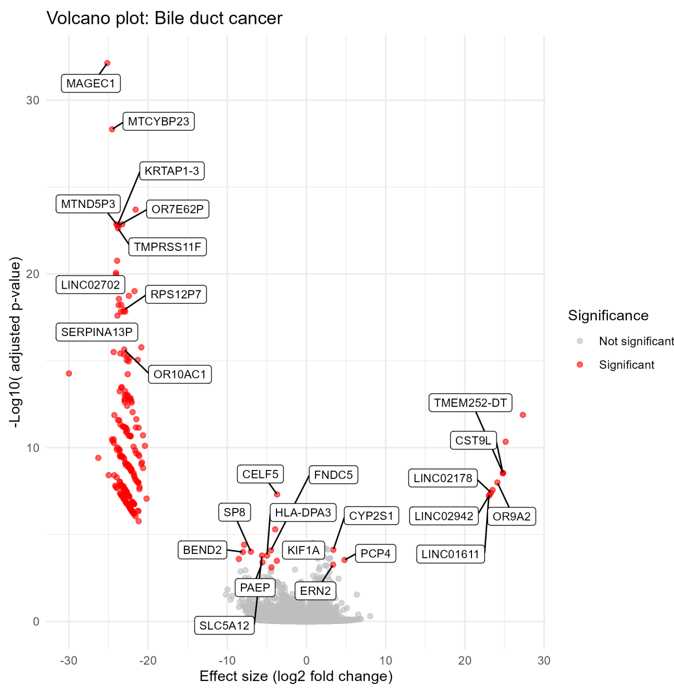

Figure 2: Differential gene expression in bile duct cancer patients

The plot demonstrates the differential expression that the analysis showed. Downregulated genes, with large negative log2fold changes, can be seen on the left side of the plot. Genes like: MAGEC1, MTCYBP23, KRTAP1-3, MTND5P3 and OR7E62P stand out as one of the most significantly undreexpressed genes.

On the other hand, the right side of the plot shows upregulated genes. Genes like: TMEM252-DT, CST9L and OR9A2 show a large positive log2foldchange making them one of the most significantly overexpressed genes.

Overall, a strong asymetry is visible on the plot suggesting that bile duct cancer can be characterised by widespread supression of specific gene sets and selective activation of smaller number of genes.

3.3 Enrichment analysis

Functional enrichment analysis using the KEGG database gave a limited biological signal, identifying "Olfactory transduction" as the only significantly enriched pathway. This pathway is enriched by a cluster of six genes, some of them being OR7E62P, OR10AC1 and OR9A2. These results meet the statistical cutoffs, however the presence of olfactory receptors which are typically not expressed in biliary tissue, suggests that the detected signal might have been influenced by low-count variance and high dispersion in the raw sequencing data. The data for bile duct cancer itself is small compared to some other TCGA cancers(lke breast or lung) which also contributes to the low signal.

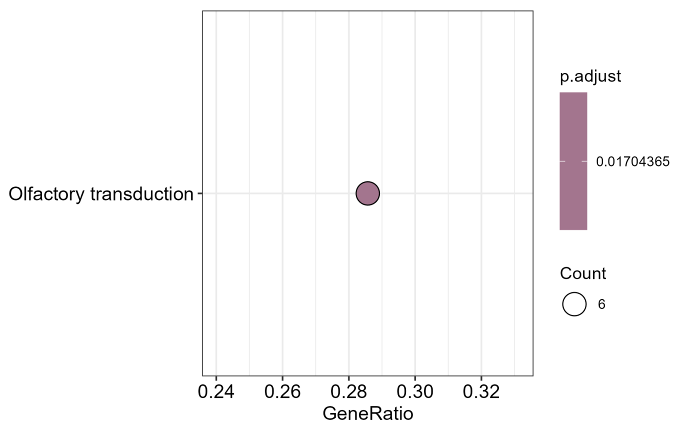

Figure 3: Enrichment analysis results

4 Discussion

This analysis of TCGA bile duct cancer dataset compared transcriptomic profiles between ealy and late-stage patients. Analysing in total 58037 genes, 250 differentialy expressed genes were identified, of which 224 were underexpressed and 26 overexpressed. Functional enrichment analysis showed that "Olfactory transduction" is the dominant biological signal. Since these genes are typically not present in the biliary tissue, this is concluded to be a tehnical artefat coming from low-count variance and high disperison in the raw data. Despite trying to loosen the statistical thresholds in attempt to capture more significant pathways, the results remained the same. This indicates that the data could lack stronger oncogenic signals.
Finally, the analysis sucessfully identified statistically significant genes that are differentially expressed. Additionally, some advanced filtering methods could be needed to move past transcriptomic noise and discover more meaningful pathways.

5 Bibliography

Juan Ramón González' material available at GitHub and CIBEREHD Bioinformatics Initiation course by Ana Maria Corraliza and Marta Coronado Zamora

Anders, S., Huber, W. (2010). Differential expression analysis for sequence count data. Genome Biology,11(10), R106.

Subramanian, A., et al. (2005). Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. PNAS, 102(43), 15545-15550.

Massberg, D., Hatt, H. (2018). The role of olfactory receptors in cancer: a review. Biological Reviews, 93(3), 1548-1563.

6 Appendix

The code used to preform the analysis is avaliable on github in the repository: $https://github.com/Nunuu8/TAB_grpu2$ under P4PetraDobric.

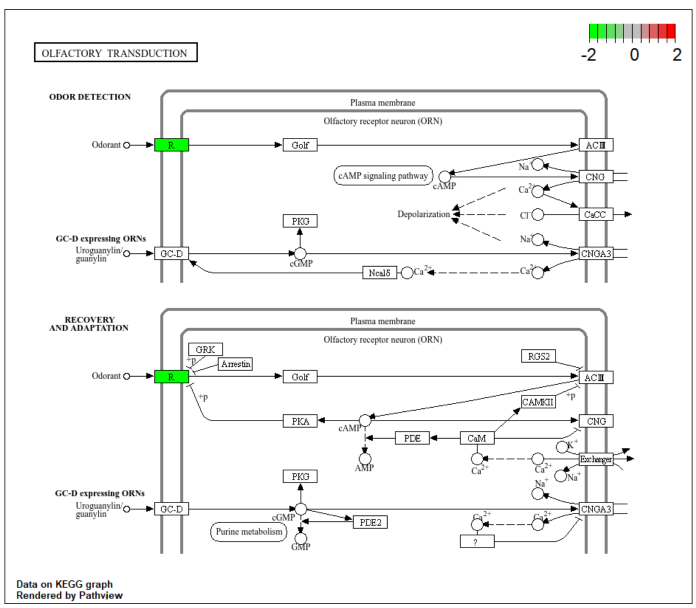

Figure 4: Suplementary figure